REPRODUCTIVE STRATEGIES

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REPRODUCTIVE STRATEGIES

- This can simply be said to be a way in which an organism reproduces or raises offsprings.
- This may involve the following; Set of behavioural, morphological and physiological adaptations that enable contact to probable mates.
- Additionally, expand the chances of mating and fertilization and improve infant continued existence.

REPRODUCTIVE STRATEGIES

Topics of discussion

1. Why is important from an aquaculture perspective to know or study reproductive diversity?
2. Differentiate asexual and sexual reproduction

REPRODUCTIVE STRATEGIES

Asexual reproduction

This involves producing of offspring that are genetically identical to a single parent and does not require fusion of sperm and egg and depends on mitotic division.

Sexual reproduction

This involves producing of offspring by fusion of haploid gametes (egg and sperm) hence needs two parents and depends on both mitotic and meiotic division.

ASEXUAL REPRODUCTION

- **Advantages**
- Fast and energy efficient as only one parent is needed
- No need to find a mate.
- When conditions in the environment allow, this can lead to a rapid increase in population

Disadvantages

- Can not bring about genetic diversity in a population.
- Changes in the environment can make the species to be vulnerable and therefore, only one habitat would be suitable to the species.
- Species may under go extinction in the case of a diseases outbreak.

ASEXUAL REPRODUCTION

Advantages

- There is genetic variation in the offspring
- Species can adapt to new environmental conditions
- A disease may not lead to an extinction in a population.

Disadvantages

- The process is quiet complicated (E.g. Finding a parner)
- Requires time and energy
- If one individual is isolated, it is impossible to increase the population since no reproduction can occur.

COMPOSITION OF FISH REPRODUCTIVE STRATEGIES

1. Gender and mating system

This is charaterised by Secondary sexual characteristics, Mating system, Sex determination and differentiation and Gender system.

2. Fish spawning

This is charaterised by number of breeding opportunities, spawning time, type of spwning and size and age at first maturation.

3. Egg fertilization and embryo development

This is charaterised by place of fertilisation, parental care, spawning site prepartion and embryonic development.

1. Gender and mating system

- Some fish species are Gonochoristic (separate sex and is fixed when the fish matures) while other are Hermaphrodictic (sex may change when the fish matures).
- However, majority of the fish species are Gonochoristic.
- About 99% of all vertebrate species (>60,000) consist of gonochorists.
- The other 1% are hermaphroditic, and almost all of them are fishes.

1. Gender and mating system

HERMAPHRODITISM IN FISH

- Simultaneous or synchronous (SH): produce both mature eggs and sperms in their gonads at the same time but self-fertilization is rare. E.g *Hypoplectrus unicolor*
- Protogyny (PG):female to male sex change (most common type).E.g *Orange spotted grouper*
- Protandry (PA): male to female sex change. E.g *Gilthead seabream*
- Bidirectional sex change(BS) E.g *Gobiodon erythrospilus*

Hermaphrodites suffer the energetic cost of maintaining two reproductive system

1. Gender and mating system

MATING SYSTEM

Monogamous: Partners live and mate only with each other.

(e.g. American freshwater catfish, some chiclids, many butterfly fish)

Promiscuous: Both sexes mate with multiple partners observed in mass spawning (e.g., Herring, cod, sea bass)

Polygynous: one male mates with multiple females (mainly freshwater fish: cichlids, African catfish, arapaima but also seawater species: sculpins (Cottoidea), sunfishes)

Polyandrous: Each female mate with several males. In this system, the male may assume the parental role (e.g., seahorse, green swordtails and Trinidadian guppies)

1. Gender and mating system

Secondary sexual characteristics

Any physical characteristic developing at puberty which distinguishes between the sexes but is not directly involved in reproduction.

2. FISH SPAWNING

NUMBER OF BREEDING OPPORTUNITIES

Semelparous or monocyclic:Spawn once and die. e.g., Pacific salmons (*Onchorynchus*sp.), eels (*Anguilla*sp.)

Iteroparaousor polycyclic: Multiple spawners

Total spawner: one spawning during spawning period (e.g. Brown trout)

Batch spawner: release multiple batches of eggs per breeding season (e.g., Atlantic cod, Halibut, few sea bass; many flatfish; seabream)

2. FISH SPAWNING

NUMBER OF BREEDING OPPORTUNITIES

Determinate batch spawners: known number of eggs prior to spawning (generally cold water species)

Indeterminate batch spawners: eggs to be spawned are not all present as oocytes in ovary prior to spawning (some develop later) (generally warm water species)

2. FISH SPAWNING

AGE AT FIRST MATURATION

Species	Female	Male
Rainbow Trout	2-3 y	1 y
Common Carp	< 1 y	< 1 y
Atlantic Cod	2 y	2 y
Sea bass	2-3 y	1 y
Atlantic Halibut	6 y	3 y
Nile Tilapia	4-6 month	4-6 month

2. FISH SPAWNING

TYPE OF SPAWNING

❑ Open substrate spawner:

1. Pelagic spawners
2. Benthic spawners

➢ Spawners on coarse bottom (rocks, gravel)

➢ Pelagic free-embryo and larvae

➢ Benthic free-embryo and larvae

➢ Spawners on plants

2. FISH SPAWNING

TYPE OF SPAWNING

- **Brood hiders**
  - Benthic spawners
  - Cave spawners
  - Spawners on/in invertebrates
  - Beach spawners

2. FISH SPAWNING

TYPE OF SPAWNING

- ❑ Nest spawners (rock, sand, plant, bubble, hole)
- ❑ Bearers
  - ➢ External bearer (transfer, forehead, mouth, skin brooders)
  - ➢ Internal bearers (ovoviviparous, viviparous)

3. EGG FERTILIZATION AND DEVELOPMENT

❑ Spawning site preparations

- ➢ No preparation: most species of broadcast spawners
- ➢ Site prepared and defended: salmons, stickleback

❑ Place of fertilisation

- ➢ External: most species
- ➢ Internal:Elasmobranches, Sebastesp.

3. EGG FERTILIZATION AND DEVELOPMENT

- Embryonic development

1. Oviparity

- Animals that lay eggs
- Fertilisation is external
- Embryo develops outside the ovary
- Most invertebrates, many vertebrates, e g most reptiles, birds, fish

3. EGG FERTILIZATION AND DEVELOPMENT

❑ Embryonic development

2. Ovoviviparity(“egg-livebirth”):

- Animals retain the eggs within their body and embryonic development occurs inside.
- Fertilisationisinternal
- Eggsarenourishedbyeggyolk
- Somefish(e.g.makosharks,sandtigersharks,seahorse),reptilesandamphibians



Ovoviviparous seahorse

3. EGG FERTILIZATION AND DEVELOPMENT

- Embryonic development

3.Viviparity(“livebirth”):

Embryo develops inside mother’sbody

Nourishment comes directly from mother

Mammals,some reptiles,some fish (e.g.,hammerheadsharks,*Xenotocaeiseni)*

3. EGG FERTILIZATION AND DEVELOPMENT

❑ Parental care

- Fish have collective pelagic spawning without any egg caring to sophisticated sexual behaviour (nest building) followed by parental care.
- Pelagic spawning and mouth brooding, nest building

ANSWER

What is the Importance of studying reproductive diversity in aquaculture context?

What is the Importance of studying reproductive diversity in aquaculture context?

- Key to our ability to manipulate reproduction of fish in captivity and practice good broodstock management
- Important for conservation of wild population

END

THANK YOU FOR YOUR ATTENTION